```
error: called from `pkg>copy_files' in file /usr/share/octave/3.2.3/m/pkg/pkg.
m near line 1473, column 13
error: called from:
error:   /usr/share/octave/3.2.3/m/pkg/pkg.m at line 756, column 5
error:   /usr/share/octave/3.2.3/m/pkg/pkg.m at line 287, column 7
```

The error that you see above is because the package installation is run as a normal user account, which has no write permissions to the */usr/lib/** directories (the default location at which the packages are installed). To avoid this, tell Octave to install the package to another location using the *pkg prefix* command:

```
octave-3.2.3:17> pkg prefix ~/oct-packages ~/oct-packages/arch
```

In the above example, I have created a directory oct-packages in my user account's home directory, and am telling Octave to install packages there. The second directory (which is a sub-directory arch under the previous directory) specifies the location where architecture-specific Octave packages should be installed.

Now, let us proceed with the install. The *-local* switch is added to specify that the package should be installed for the current user only. If you want a global installation, please omit it:

```
octave-3.2.3:18> pkg -local install sockets-1.0.6.tar.gz

warning: gen\_doc\_cache: unusable help text in 'sockets'. Ignoring function.
```

You can ignore the warning for now, since it relates to the help text. Let us try some functions from the *sockets* package:

```
octave:2> s=socket();
octave:3> serverinfo.addr="www.google.com";
octave:4> serverinfo.port=80
serverinfo =
{
  addr = http://www.google.com
  port =  80
}

octave:5> connect(s,serverinfo)
ans = 0
```

So, as you can see, we have successfully connected to the given server at the specified port. As an exercise, you should try to write a simple client-server code in Octave using these simple function calls. The functions in the sockets package are described at *http://octave.sourceforge.net/sockets/overview.html*

Creating an Octave package

Creating an Octave package is a fairly elaborate process, and something you should attempt only after you have gained a certain familiarity with Octave. Once you have your package ready, you will need to push it to the relevant place in the Octave-Forge project, to make it available to the community.

Compatibility with MATLAB

As stated in the first article in this series, Octave was built with MATLAB compatibility in mind. I have no extensive experience in working with either, but the general opinion seems to be that if you are targeting both, your best bet is to use Octave and hope that it works with MATLAB too. MATLAB's parser, for example, has issues working with perfectly legal Octave scripts. See *http://wiki.octave.org/wiki.pl?MatlabOctaveCompatibility* for details on this.

GUI and Web interfaces

Although you would generally like the no-frills command-line interface of Octave, there are some GUI and Web-based options to try out as well. QtOctave (*http://qtoctave.wordpress.com/*) is a Qt front-end for Octave, and is very actively being developed. If you use KDE, trying this should be easy.

There is also a Web interface to Octave (*http://hara.mimuw.edu.pl/weboctave/web/*), which is useful to try out Octave even without installing it.

So, this is a rather short culmination to the series on Octave packages, and how they extend Octave's core functionalities. Octave is just one of the many open source tools for research and academic work. My aim in this series on Octave was to bring to light this open source alternative to MATLAB, and encourage its use among students and teachers alike. I hope you have appreciated the series, and will try to incorporate your newly acquired knowledge of Octave in your tasks. END

Resources

- GNU Octave: *http://www.gnu.org/software/octave/*
- GNU Octave Documentation: *http://www.gnu.org/software/octave/docs.html*
- Creating Octave packages: *http://www.network-theory.co.uk/docs/octave3/octave_281.html*
- Octave-Forge developers guide: *http://octave.sourceforge.net/developers.html*
- 5.MATLAB-Octave compatibility: *http://www.gnu.org/software/octave/FAQ.html#MATLAB-compatibility*
- 6.My article, "Hey, Researcher! Open Source Tools For You!" (http://*linuxgazette.net/174/saha.html), talks about a number of other open source scientific and numerical tools.*

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